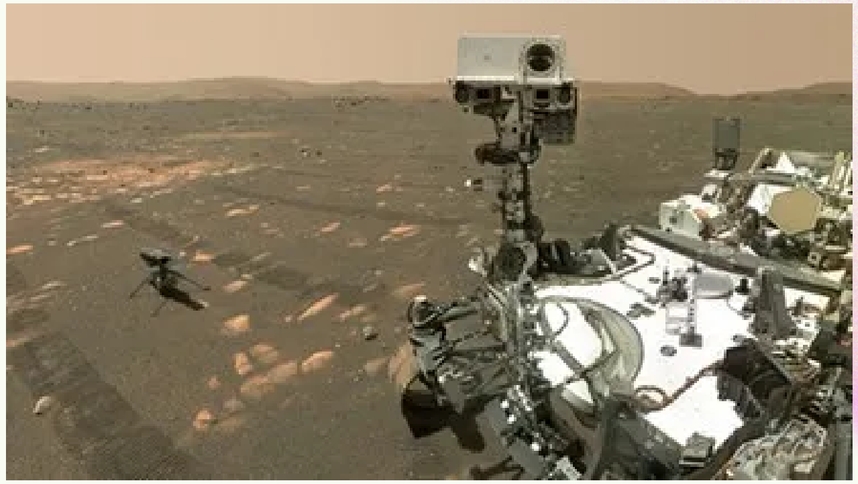
The Architecture of Mars Global Localization for the Perseverance Rover: Onboard Map Matching and Hardware Repurposing



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Introduction

- The Mars rover traditionally relied on incremental odometry and periodic localization updates from Earth for its navigation.
- The errors keep accumulating for every estimation and this causes delays and constraints on the autonomous mobility of the rover because of communication delays.
- To solve this issue, the Mars Global Localization (MGL) system was implemented, which improved not just the navigation systems but also the operational efficiency and overall mission capability

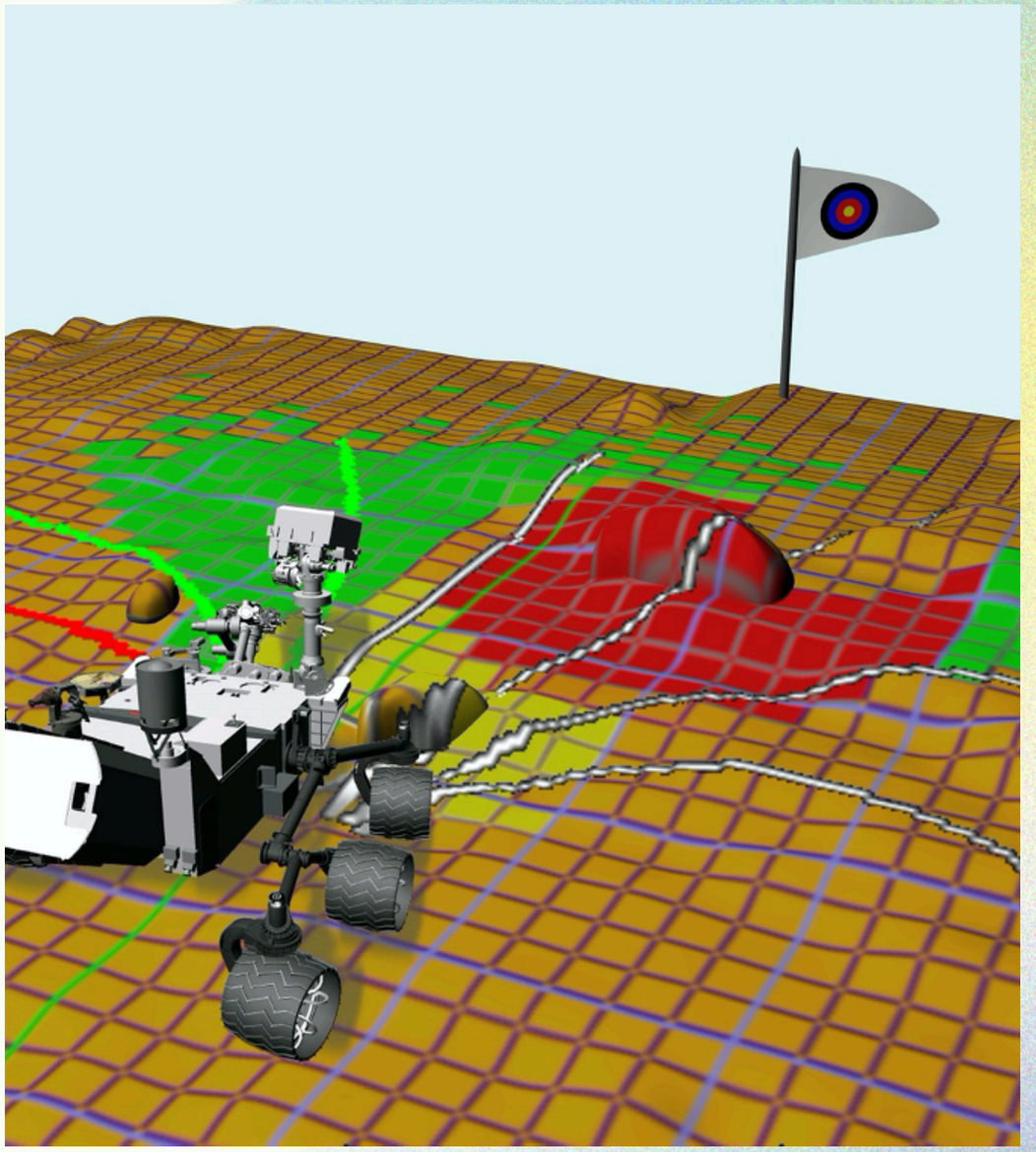


Fig: Graphical representation of the rover's navigation system

The Problem: Odometry Drift and the "Ground-in-the-Loop" Delay

- Dead Reckoning (Sojourner): Early rovers tracked wheel rotations to estimate position. However, loose Martian sand causes wheel slippage, leading to compounded positional errors over time (Odometry Drift).
- The Visual Odometry (VO) Upgrade: Later rovers (MER, Curiosity) improved accuracy by comparing consecutive stereo camera images to track physical landmarks.
- The Trade-off: While VO significantly reduced drift, it was computationally heavy and drastically reduced early driving speeds.

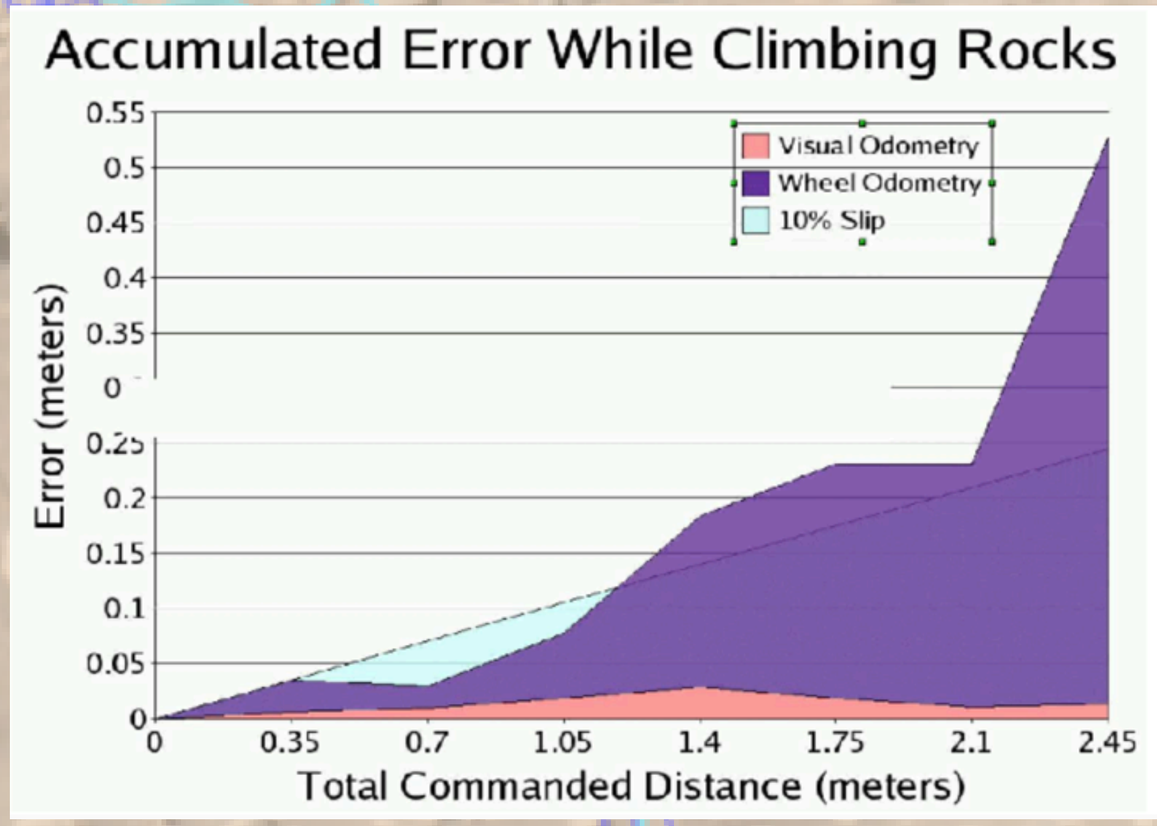


Fig. Visual Odometry Error measured during a 2.45 meter drive using HAZCAMs on the MER Surface System Testbed Lite rover.

Fig : Perseverance's path through a dense section of boulders (light blue- center of wheel hubs, dark blue path of six wheel

Perseverance and the 24-Hour Delay

- Persistent Drift: Despite advanced VO, Perseverance still accumulates tiny positional errors on long drives (up to 35 meters).
- Safety Halts: When positional uncertainty grows too high, the rover's safety systems trigger a mandatory halt.
- The Manual Bottleneck: The rover beams a 360° panorama to Earth.
- Human experts manually cross-reference the images with satellite maps.
- New coordinates are sent back to Mars.
- The Result: This "ground-in-the-loop" process takes a full day or more, severely limiting exploration speed and necessitating an autonomous onboard solution.

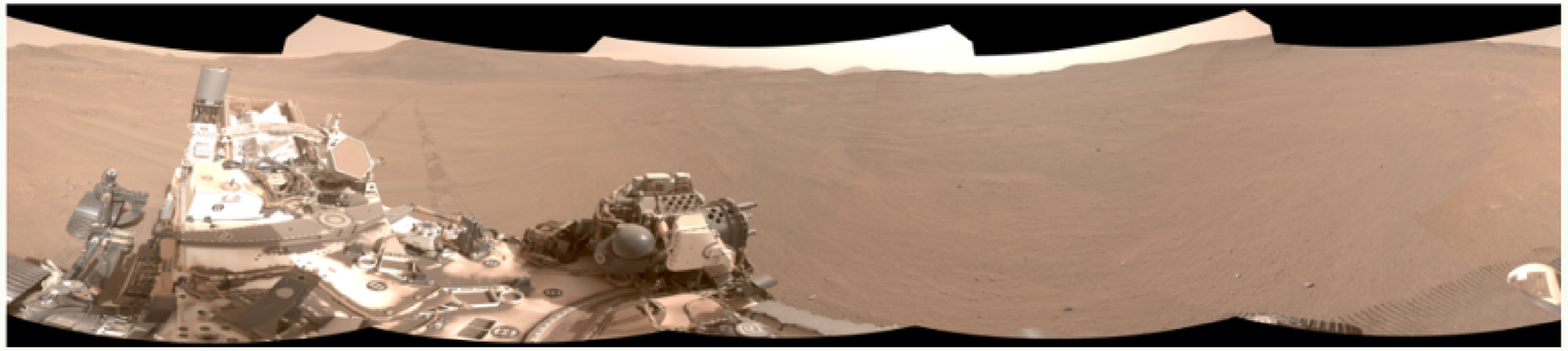


Fig. - Panorama from Perseverance is composed of five stereo pairs



Fig. – Onboard Mars Global Localization algorithm used to pinpoint location that matches terrain features in navigation camera shots

The Hardware: Repurposing Ingenuity's Brain

- For Mars Global Localization to work, Perseverance needs to process large amounts of image data and compare rover images with orbital maps. This requires significant computational power.
- The RAD750 it can survive the harsh environment of Mars, but it is not ideal for running modern computer vision algorithms. However, because it was designed for durability rather than performance.

Feature	RAD 750
Type Clock Speed Performance Power Usage	Radiation-hardened processor 110-200 MHz 266 MIPS+ About 5 W

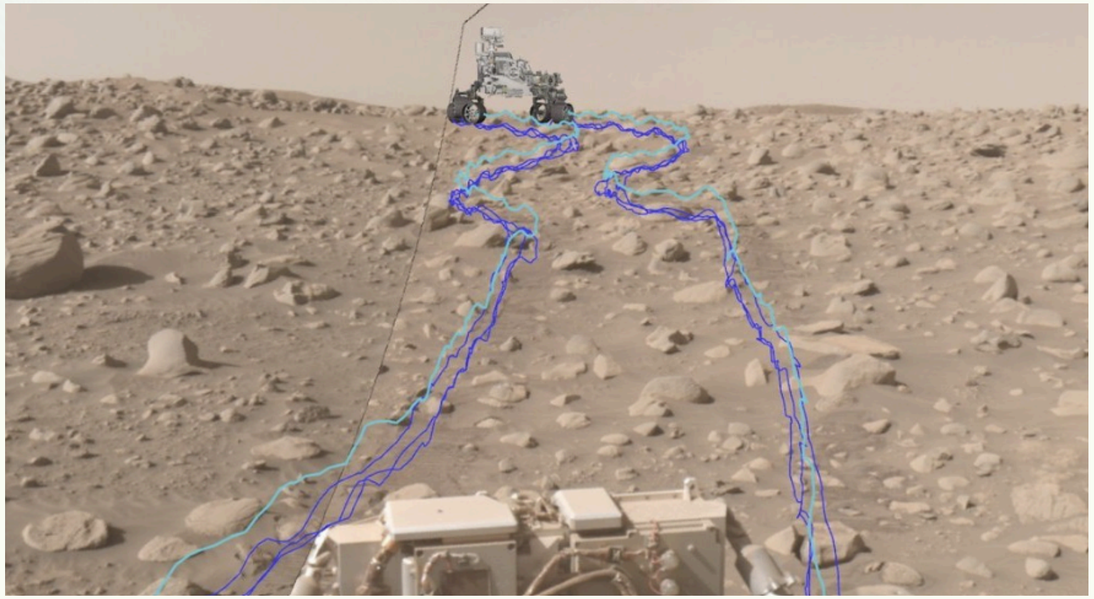


Fig: The Perseverance Rover following a path planned by the MGL system

Perseverance is the most autonomous Mars rover ever built. It has achieved record-breaking autonomous drives of 699.9m over three days which is a planetary rover record.

Computational Requirements of Localization

Rover localization is modeled as an image registration problem between the rover images and the orbital map.

- Absolute Orientation Estimation:

Rover missions can typically estimate absolute orientation with high accuracy, either by estimating rover orientation based on the position of the sun, the stars, or internally through gyrocompassing.

- Panoramic Stereo Imaging:

A 360 panorama of stereo pairs around the rover is acquired to maximize the area of imaged terrain.

Repurposing Ingenuity’s Processor

- The fast processor on the Ingenuity Helicopter Base is mounted on the Perseverance rover, to perform the localization. The Ingenuity system utilizes a commercial Qualcomm Snapdragon 801 processor.

- This was the first time a non-radiation-hardened processor was used as a computing unit on another planetary surface mission.



Fig: The Ingenuity Helicopter - a visualisation

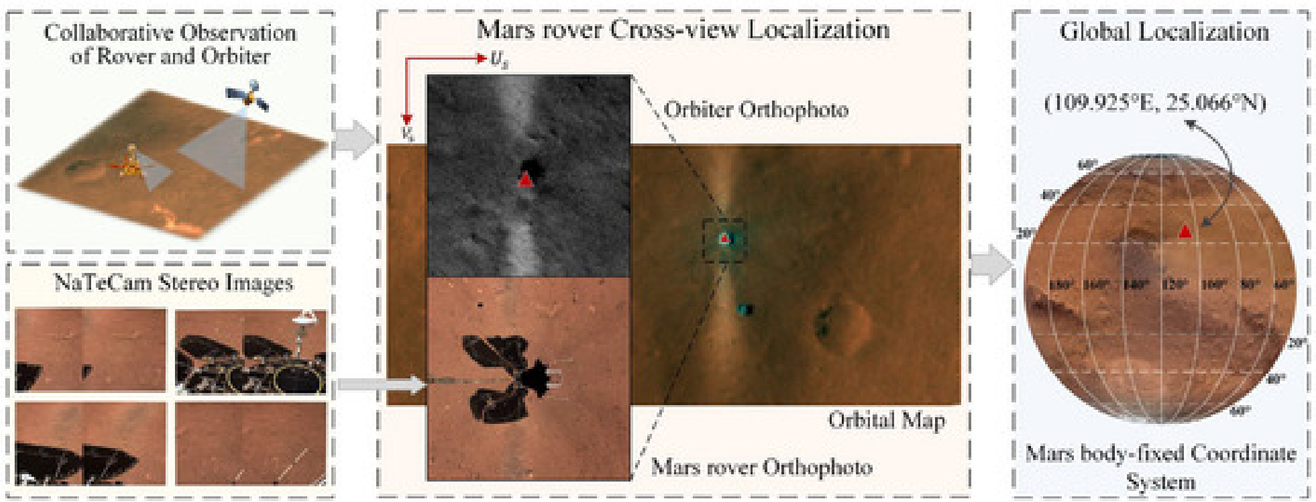
The Software behind MGL - How the "censible" framework works

The Three-Step main steps

1. **IMAGING**
   - Panoramic Imaging &
   - Radiometric Correction

2. **PROJECTION**
   - Top-Down Orthomosaic
   - Projection

3. **MATCHING**
   - Cross-Referencing with Orbital Maps:



Autonomous processing resets position uncertainty within 2 minutes without Earth-based communication.

Results on the Perseverance Rover

This framework was implemented on Mars on the Perseverance rover and successfully executed on Martian sol 914.

Algorithm	Avg. Error	99%-Tile	>5m Errors
Masked NCC	24.97m	45.90m	225/264
Masked Mutual Information	24.33m	46.98m	216/264
Modified Census (Ours)	**0.36m**	**0.93m**	**0/264**

Table 1: Comparison of similarity measures, demonstrating the sub-meter accuracy of the chosen Modified Census algorithm

Resolution	Pairs	Total	Stereo	Matching	Memory
2560x1920	1	36s	14s	8s	<1GB
2560x1920	5	81s	65s	8s	<1GB

Table 2: Execution timing on flight-identical hardware, showing the algorithm completes within two minutes using minimal volatile memory

Conclusion

The Perseverance rover is now capable of navigating the Martian surface with unprecedented independence.

The re-purposing of the Ingenuity Helicopter Base Station provided the necessary computational bandwidth.

This framework provides human level accuracy

This framework opens up a world of possibilities for future rovers

Essential for missions with massive communication delays (e.g., icy moons of Europa and Titan)

Proves software architecture is now as critical as mechanical engineering in deep-space exploration.

Thank you!

Do you have any questions?